

KONE Monospace: Economic Value to Consumer

Summary:

We will use the Monospace case to illustrate EVC. To focus primarily on EVC, a quick summary of relevant case facts is provided below (for ease of comparison, base your computations on the numbers **below**). As the case notes, considerable debate exists within KONE about the willingness-to-pay of various customers for the new elevator, and the right price to set. Your task consists of computing the EVC for MonoSpace, and constructing an EVC demand curve for launch planning purposes.

All elevators required a machine room to house the elevator equipment. The cost of incorporating a machine room typically depended on where it was placed within a building. For instance, machine rooms for geared traction-based elevators could be placed either on top of the shaft (PT type), or at the base of the shaft (PU type); those for hydraulic elevators (PH type) were typically placed on the lowest floor (see Exhibit 2 for common arrangements). Based on case numbers, the average potential market of 11,500 elevators can be broken down as follows: 60% PH type, 27% PU, and 13% PT.

Table 1 (based on case data) shows the current prices and operating costs of the elevators.

Table 1: prices and operating costs of the elevators.

	PH	PT	PU
Price (DM)	60,000	75,000	80,000
Energy consumption (DM per year)	7200	5000	5000
Oil requirements (DM per year)	1000	100	100
Fuses (DM per year)	1600	800	800

The key innovation of the MonoSpace elevator was the use of a proprietary induction-motor based technology called “EcoDisc,” which eliminated the need for a machine room. The cost of a machine room was estimated to be approximately 1/4 the cost of installing and purchasing an elevator. The MonoSpace also offered better ride-comfort (i.e. was faster), required less electricity (DM 3000 per year), entailed lesser fuse costs (DM 60 annually) and required no oil (eliminating potential fire and environmental concerns). Its installation time was only 190 hrs, 60 hrs less than the other elevators in the market.

Questions:

1. Compute the economic value of Monospace to potential PH, PT and PU buyers. Assume that the average life-time of an elevator is 10 years. Is there a component of EVC that is not quantifiable?
2. Plot the EVC demand curve for Monospace.
3. Assume that KONE spent 10 million DM in developing the Monospace technology, and that current marginal cost per unit of MonoSpace is DM 65,200. Plot the profits for KONE as a function of prices, and determine the optimal price. Compute the optimal price assuming the cost of developing the Monospace technology was a) 30 M and b) 80 M.